

Logic Design and Switching Theory

Lecture – 3

Subtitle: Introduction to Combinational Logic Circuit
and Design

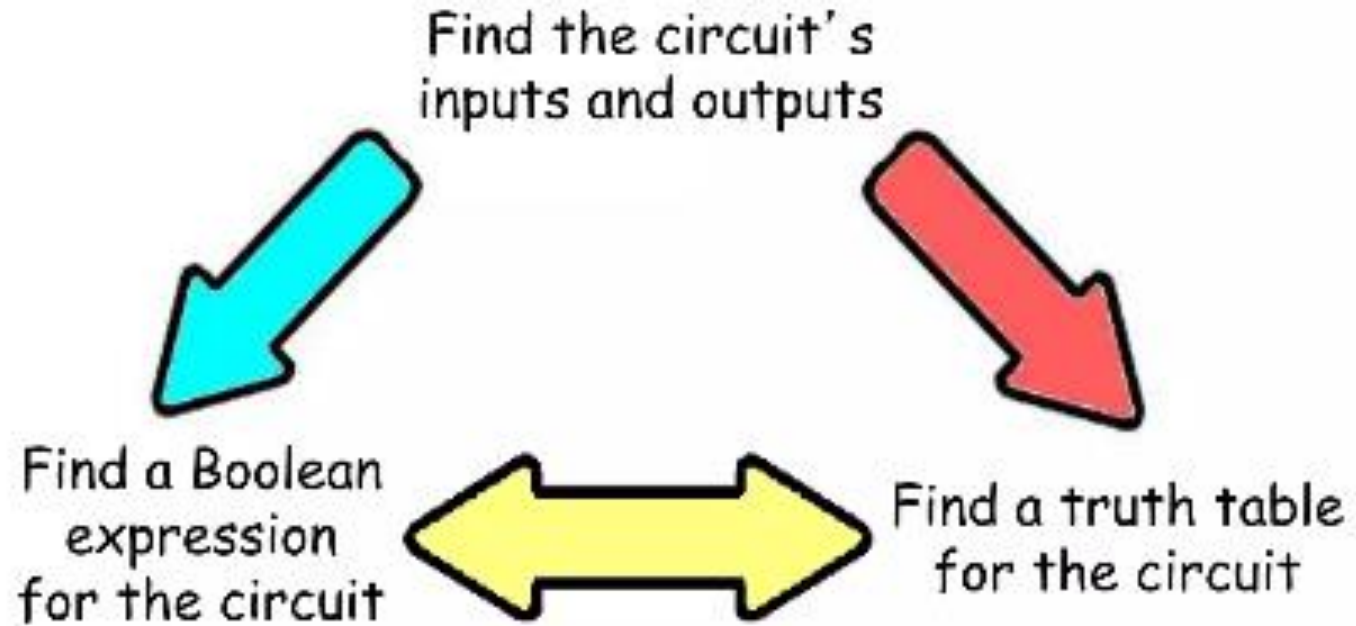
by
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LOs – Learning Objectives

**Previous
ICs
History of ICs
Digital Circuits**

Combinational Circuit

Previous...



Previous...

	A B C	Minterm	
0	0 0 0	m_0	$\overline{A} \overline{B} \overline{C}$
1	0 0 1	m_1	$\overline{A} \overline{B} C$
2	0 1 0	m_2	$\overline{A} B \overline{C}$
3	0 1 1	m_3	$\overline{A} B C$
4	1 0 0	m_4	$A \overline{B} \overline{C}$
5	1 0 1	m_5	$A \overline{B} C$
6	1 1 0	m_6	$A B \overline{C}$
7	1 1 1	m_7	$A B C$

	A B C	Maxterm	
0	0 0 0	M_0	$A + B + C$
1	0 0 1	M_1	$A + B + \overline{C}$
2	0 1 0	M_2	$A + \overline{B} + C$
3	0 1 1	M_3	$A + \overline{B} + \overline{C}$
4	1 0 0	M_4	$\overline{A} + B + C$
5	1 0 1	M_5	$\overline{A} + B + \overline{C}$
6	1 1 0	M_6	$\overline{A} + \overline{B} + C$
7	1 1 1	M_7	$\overline{A} + \overline{B} + \overline{C}$

Previous...

AB \ CD		CD			
		00	01	11	10
00	00	1 0	1 1	0 3	1 2
	01	1 4	0 5	0 7	1 6
11	11	0 12	0 13	1 15	1 14
	10	0 8	1 9	0 11	0 10

$B_3B_2 \backslash B_1B_0$		B_1B_0			
		00	01	11	10
00	0	0	1	1	
01	1	1	0	0	
11	1	1	0	0	
10	0	0	1	1	

Structure of a K-map

- K-maps are grids where each cell represents a **minterm** (binary combination of input variables).
- The number of cells equals 2^n , where n is the number of input variables:
 - ✓ 2 variables $\rightarrow$ 4 cells.
 - ✓ 3 variables $\rightarrow$ 8 cells.
 - ✓ 4 variables $\rightarrow$ 16 cells.

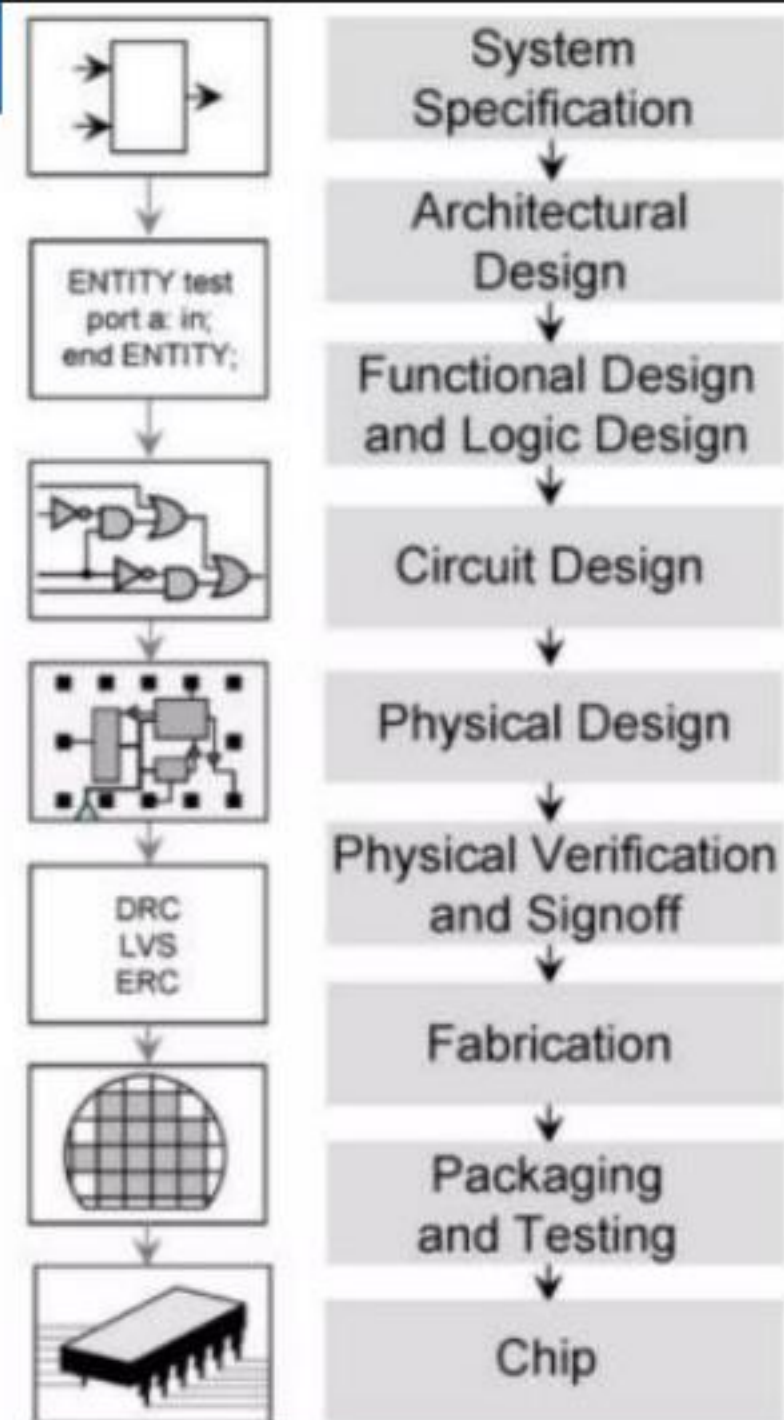
Integrated Circuits

IC is a circuit consisting of active and passive elements fabricated on a single semiconductor chip and in an individual package.

Technically chip refers to the actual piece of silicon on which the circuit is fabricated, but the term chip and fabricated are often used interchangeably.

History of Integrated Circuits

- ✓ An IC also referred as chip or micro chip is a set of electronic circuits on one small chip of semiconductor material, normally silicon.
- ✓ This can be made much smaller than a discrete circuit made from independent electronic components. IC's can be made very compact having up to several billion transistors and other electronic components in an area the size of a fingernail.
- ✓ The width of each conducting line in a circuit can be made smaller and smaller as a technology advances; in 2008 it dropped below 100 nanometers and has now been reduced to tens of nanometers.



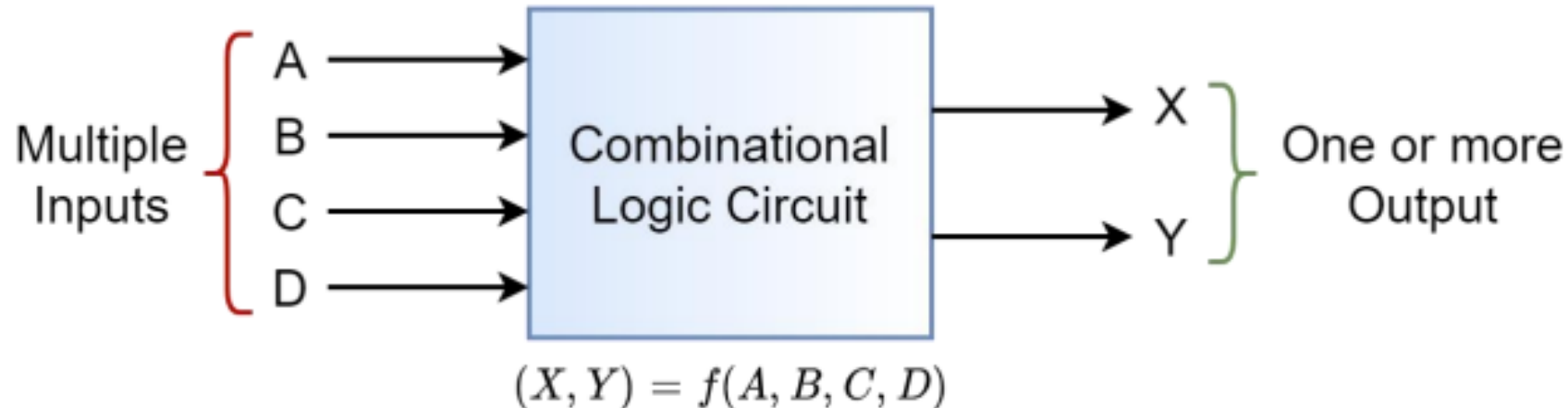
Categories of ICs

➤ Gate/transistor ratio is roughly 1/10

- SSI < 12 gates/chip
- MSI < 100 gates/chip
- LSI ...1K gates/chip
- VLSI ...10K gates/chip
- ULSI ...100K gates/chip
- GSI ...1Meg gates/chip

Digital circuits (combinational and sequential)

Digital logic circuits can be classified into “combinational” and “sequential”. A combinational logic circuit is one whose output solely depends on its current inputs. sequential circuits, on the other hand, sequential circuits are built using combinational circuits and memory elements called “flip-flops”.

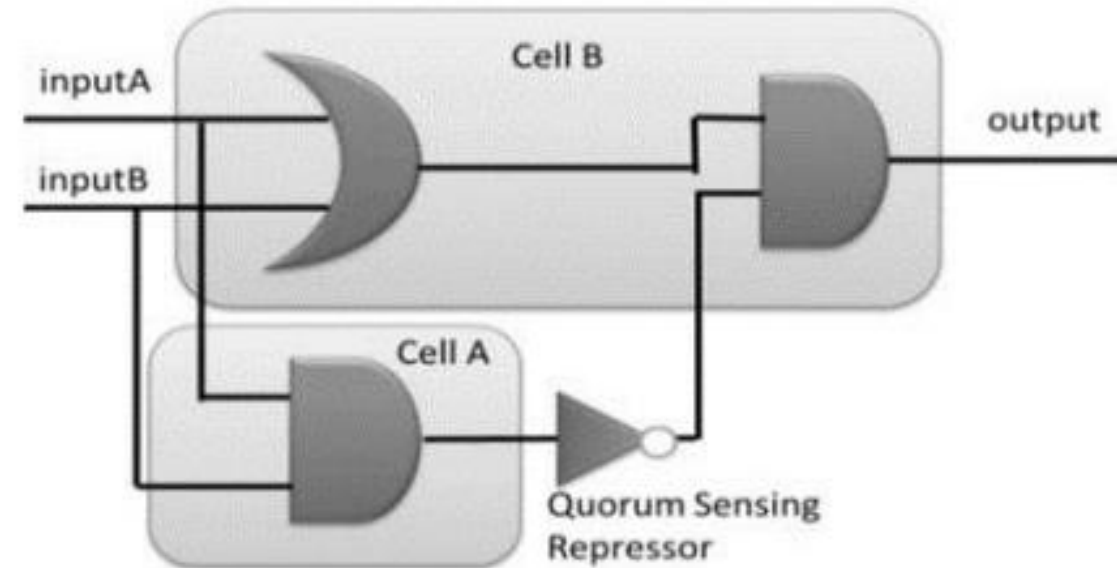


Combinational Logic Circuit & Design

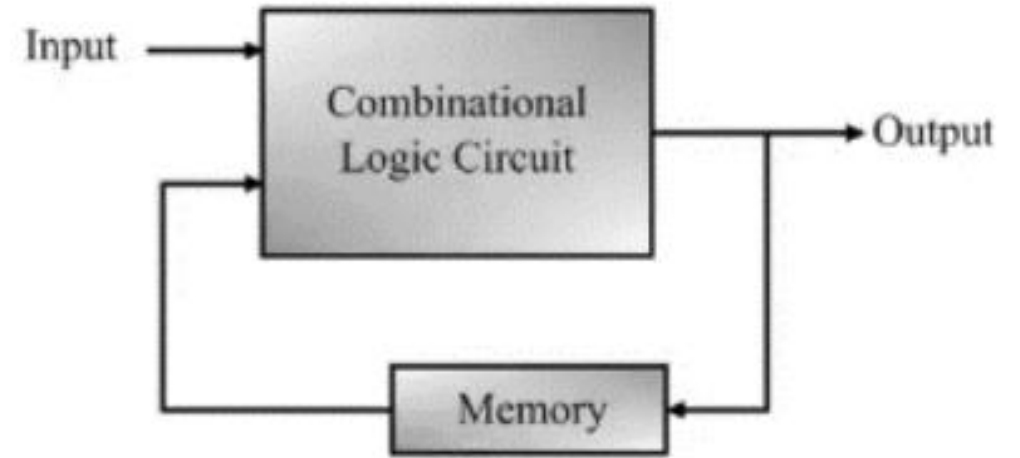
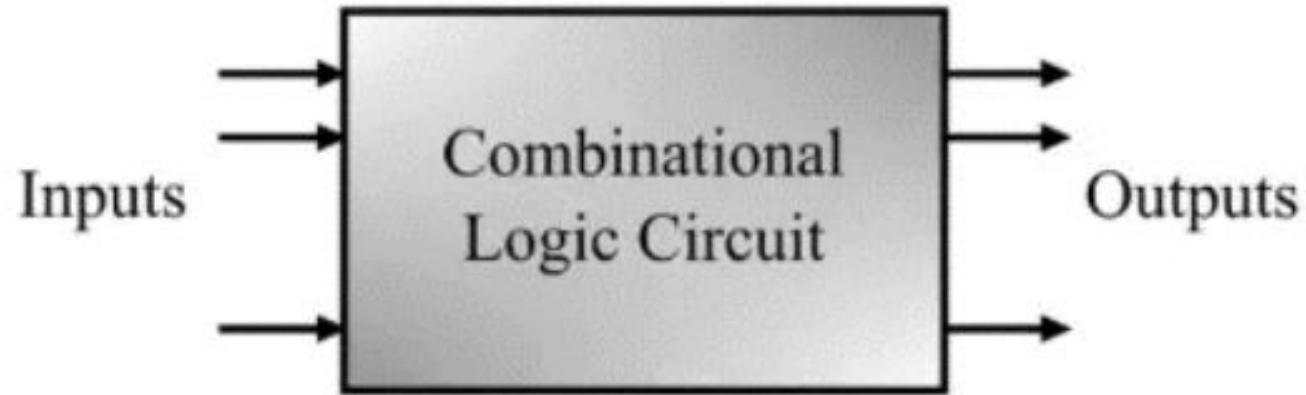
- A combinational logic circuit is one in which the present state of the combination of the logic inputs decides the output .
- The term combination logic means combining of two or more logic gates to form a required function where the output at a given time depends only on the input.

Characteristics of Combinational Logic Circuit

- The output of combinational circuit at any instant of time, depends only on the levels present at input terminals.
- The combinational circuit do not use any memory. The previous state of input does not have any effect on the present state of the circuit.
- A combinational circuit can have an n number of inputs and m number of outputs.



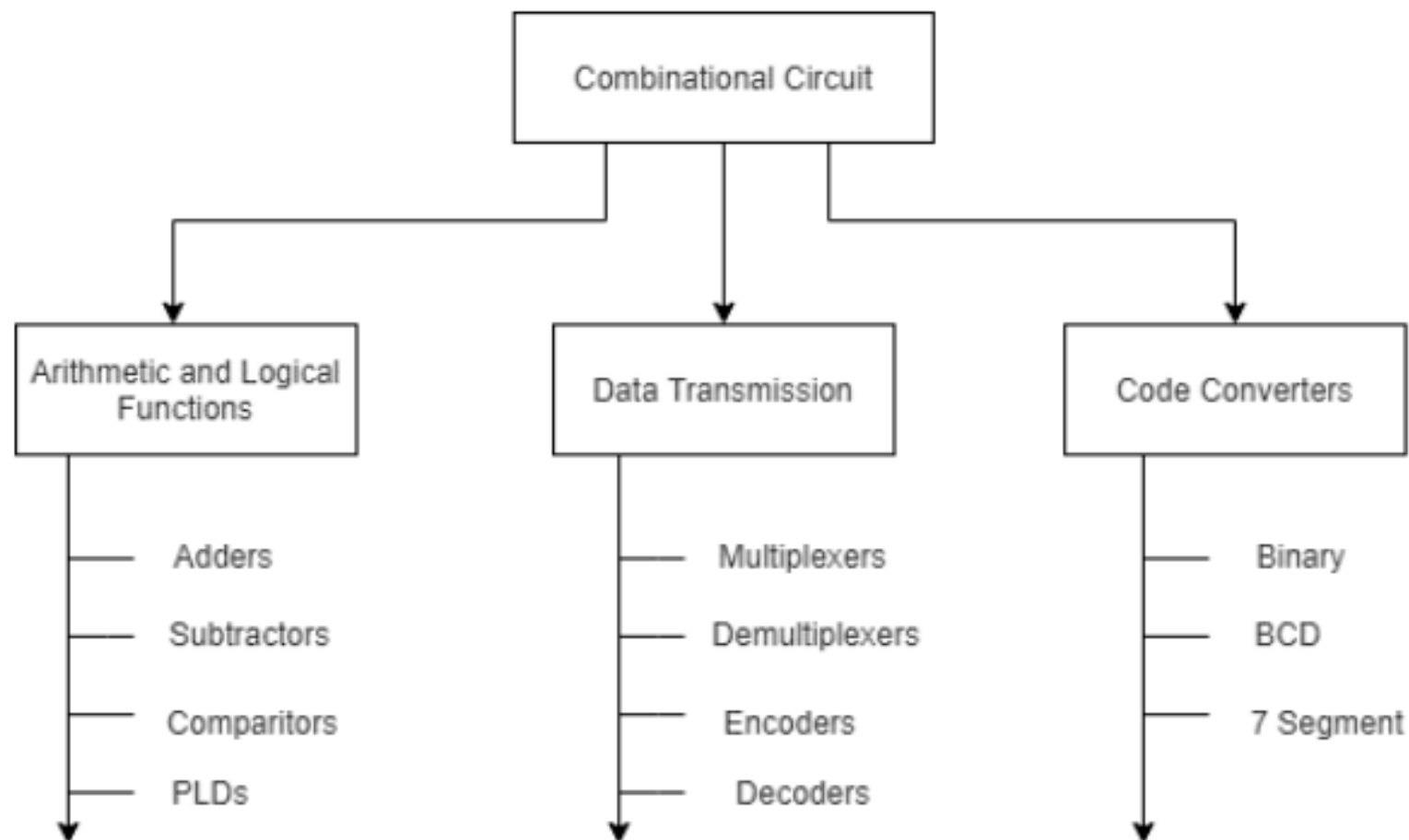
Combinational vs Sequential circuits



Difference

Feature	Combinational Circuits	Sequential Circuits
Memory	No memory	Requires memory
Output	Depends on inputs	Depends on inputs + past states
Examples	Adders, Encoders	Flip-flops, Counters

Classification



Combinational Logic Circuit & Design

Characteristics:

- No feedback loops.
- Outputs depend only on the combination of inputs

Basic Building Blocks:

- Logic Gates: AND, OR, NOT
- Derived Gates: NAND, NOR, XOR, XNOR

Used in:

- Arithmetic and logic units (ALU)
- Data transmission systems
- Digital signal processing
- Switching circuits

Combinational Logic Circuit & Design

Advantages:

- Fast and reliable
- Easier to design compared to sequential circuits.

Limitations:

- Cannot store data.
- No memory or feedback.

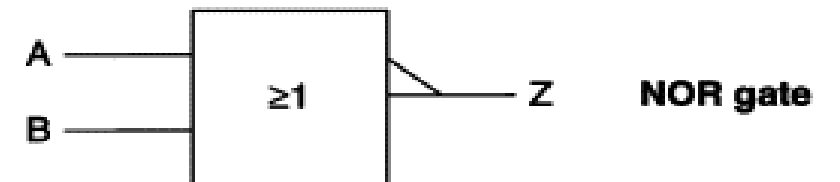
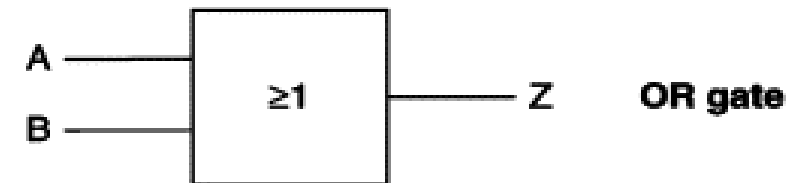
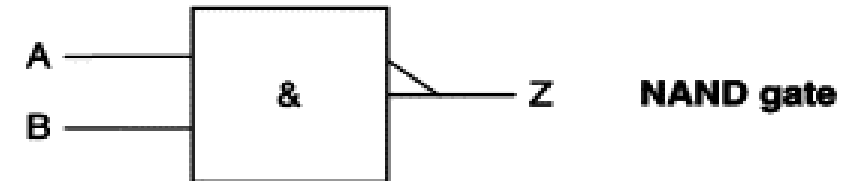
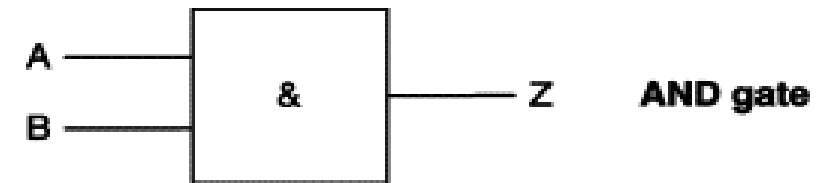
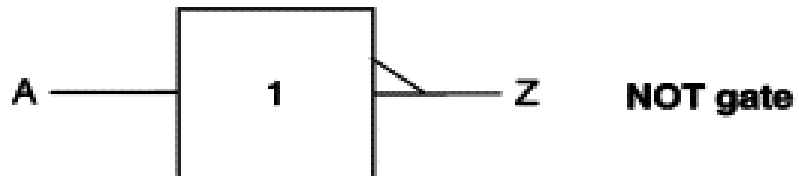
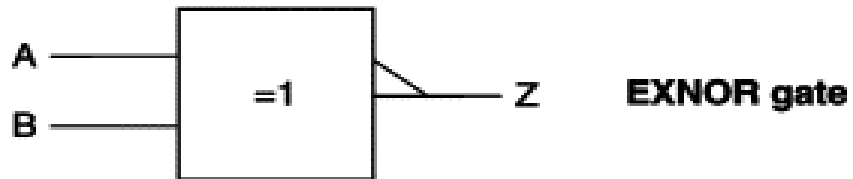
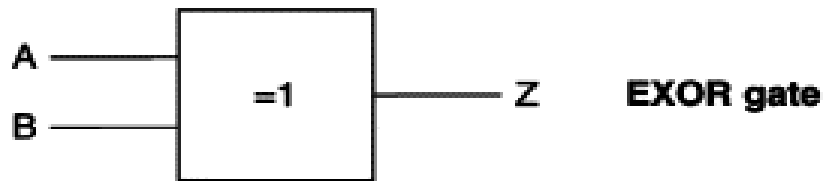
Key Features:

- Stateless.
- Real-time operation

Combinational circuits are foundational to digital electronics, enabling the design of complex systems.

Combinational Logic Circuits

An alternative set of logic symbols, IEEE/ANSI standard 91-1984 (Graphics Symbols for Logic Functions), is shown here:



Summary

- Combinational circuits produce output based only on current inputs.
- Common building blocks include logic gates, adders, multiplexers, decoders, and encoders.
- Applications range from arithmetic operations to control systems.
- Design involves truth tables, Boolean expressions, and circuit diagrams.

Thank you!